

Importance Sampling: $E_f[w(x)]$

* Examples:

$$\textcircled{1} E_f[w(x)] = \frac{1}{n} \sum_{i=1}^n w(x_i) \quad x_i \sim f(x)$$

but we cannot sample from $f(x)$.

- $\textcircled{2}$ Sometimes we may want to purposefully choose a different distribution, even if
↳ we can sample from $f(x)$.

Consider an example:

$$S = (x_1 + x_2 + \dots + x_n) \quad x_i \in \mathbb{R}$$

$$\text{attention} = a_1 v_1 + a_2 v_2 + \dots + a_n v_n \quad v_i \in \mathbb{R}^d$$

output

goal is to sample and estimate S

$$S = n E[x_i] \quad i \sim [1, \dots, n] \text{ at random.}$$

$$p(x) = \text{Uniform} \{1, \dots, n\}$$

$$\hat{S} = n \cdot \frac{1}{n} \sum_{j=1}^n x_{i_j}$$

$\textcircled{Q}$ What are the cases in which this is not desirable sampling

Example: values of x_i $\{10^6, 1, 1, 1, \dots, 1\}$
 $n = 100$

$E(\hat{S}) = S$ it is still unbiased,
but variance will be high?

Q. What should be good distribution?

→ A distribution that samples x_i with high probability or
in this particular case = 1.

* This condition is actually the case in
attention computation.

$$0 = \sum a_i \vec{v}_i \quad \|\vec{0}\| = \text{Constant},$$

$$a_i = \frac{e^{\langle q, k_i \rangle}}{\sum_j e^{\langle q, k_j \rangle}}$$

depending on
dot products
subsequent keys
far from query q
get exponentially
small!

→ Random Sampling will not work!

Formalise this idea in importance sampling.

X : Random variable

$w(x)$: function of interest

$E[w(X)]$: Quantity of interest

f
 $f(x)$: "true distribution"
 $g(x)$: proposed distribution.

$$E_f[w(x)] = E_g[\underbrace{\quad}_{?}]$$

$$\int w(x) f(x) dx = \int \frac{w(x) f(x) \cdot g(x)}{g(x)} dx$$

$$= E_g \left[\frac{w(x) f(x)}{g(x)} \right]$$

$$\underbrace{E_f[w(x)]}_{\hat{\mu}} = \frac{1}{n} \sum \frac{w(x_i) f(x_i)}{g(x_i)} \quad x_i \sim g(x)$$

$$E_g[\hat{\mu}] = \frac{1}{n} \sum_i E_g \left[\frac{w(x_i) f(x_i)}{g(x_i)} \right]$$

$$= \frac{1}{n} \cdot n \cdot E_f[w(x)]$$

$$= E_f[w(x)]$$

* Unbiased? Yes.

So this works for any proposal $g(x)$

What's a good $g(x)$?

↳ One with least variance!!

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \frac{w(x_i) f(x_i)}{g(x_i)}$$

$$\begin{aligned} \text{Var}(\hat{\mu}) &= \frac{1}{n^2} \cdot n \cdot \text{Var}_g \left(\frac{w(x_i) f(x_i)}{g(x_i)} \right) \\ &= \frac{1}{n} \left[E_g \left(\frac{w^2(x) f^2(x)}{g^2(x)} \right) - \mu^2 \right] \end{aligned}$$

Problem:

$$\begin{aligned} &\arg \min_g \underbrace{E_g \left[\frac{w^2(x) \cdot f^2(x)}{g^2(x)} \right]} \\ &\int g(x) dx = 1 \\ &g(x) > 0 \end{aligned}$$

$$\int \frac{w^2(x) f^2(x)}{g^2(x)} \cdot g(x) \cdot dx$$

$$= \int \frac{w^2(x) f^2(x)}{g(x)} dx \quad \left[\begin{array}{l} \text{Choose a } g(x) \\ \text{which} \\ \text{minimises} \\ \text{this.} \end{array} \right]$$

How do we proceed?

Simplicity Consider discrete case

$$\min_{g(x)} \sum_i \frac{\omega^2(x_i) f^2(x_i)}{g(x_i)}$$

$$\sum g(x_i) = 1$$
$$g(x_i) > 0$$

$$\sum_i \frac{\omega^2(x_i) f^2(x_i)}{(\sqrt{g(x_i)})^2} \quad \left| \quad \sum (\sqrt{g(x_i)})^2 = 1 \right.$$

If every "i" is a dimension

$$\min. \quad \left\| \frac{\omega(x_i) f(x_i)}{\sqrt{g(x_i)}} \right\|_2^2 * \underbrace{\left\| \sqrt{g(x_i)} \right\|_2^2}_{=1}$$

$$\|a\|_2^2 \|b\|_2^2 \geq ?$$

Cauchy Schwarz 2

$$\|a\|_2 \|b\|_2 \geq \langle a, b \rangle$$

There is a integral version of this

$$\begin{aligned} \int \left(\frac{|w(x)| f(x)}{\sqrt{g(x)}} \right)^2 dx &= \int (\sqrt{g(x)})^2 dx \\ &\geq \left(\int |w(x)| \cdot f(x) dx \right)^2 \\ &= \underbrace{E_f[|w(x)|]}_{\substack{\uparrow \\ \text{minimum value}}} \end{aligned}$$

is Cauchy Schwartz tight?

Yes!

When is minimum achieved?

$$\sqrt{g(x)} \propto \frac{|w(x)| f(x)}{\sqrt{g(x)}}$$

$$\boxed{g(x) \propto |w(x)| f(x)}$$

$$g(x) = \frac{|w(x)| f(x)}{\int |w(u)| f(u) dx}$$

$$g(x) = \frac{|w(x)| f(x)}{E_f[|w(x)|]}$$

This is the best proposal distribution.

Does it make sense?

$|w(x) \cdot f(x)|$ is exactly the "Contribution" of i^{th} term to the expectation.

$w(x) > 0 \forall x$, then what is the variance?

$$g(x) = \frac{w(x) f(x)}{E_f(w(x))}$$

$$\text{Var}(\hat{\mu}) = \frac{1}{n} \left[\int \frac{w^2 f^2}{g^2} g dx - E_f(w(x))^2 \right]$$

$$= \frac{1}{n} \left[E_f(w(x))^2 - E_f(w(x))^2 \right]$$

$$= 0 \quad !!$$

Something Wrong?

No. $g(x)$ needs you to know $E_f(w(x))$!

In practice, we will be trying to cheaply
get a proposal $g(x)$ that is
close to $f(x)/w(x)$.

Attention example

$$o = \sum_{i=1}^n a_i v_i$$

How should we sample terms?

Best distribution =

$$g(x) = [g(1) \dots g(n)]$$

$$g(i) \propto a_i \|v_i\|_2 \quad [\text{Exercise}]$$

a_i Computation will need entire KV
Cache reading.

↳ What is a good $g(x)$? [Think]
that is cheaply computable

we will answer this question partially
in the seminar and hopefully in
some future class.
